Natural Vegetation and Wildlife

The big idea

India is one of the **12 mega bio-diversity countries** of the world, with about **47,000 plant species** (10th in the world, 4th in Asia) and about **90,000 animal species**. **Natural vegetation** means plant communities that grow naturally without human help and have been left undisturbed for a long time; this is called **virgin vegetation**. Its type depends on climate, soil and relief.

KEY TERMS

Flora: plants of a region or period. **Fauna:** species of animals. **Endemic species:** purely Indian species found only here. **Exotic species:** plants which came from outside India. **Ecosystem:** all plants and animals in an area, interdependent and connected with their physical environment. **Biome:** a very large ecosystem on land with distinct vegetation and animal life.

Factors affecting vegetation

- **Relief: land** (fertile plains for agriculture, rugged terrain for grassland and forests) and **soil** (sandy desert soil supports cactus and thorny bushes; wet marshy deltaic soil supports mangroves; hill slopes with deep soil have conifers).
- **Climate: temperature** (falls with height, changing vegetation from tropical to sub-tropical, temperate and alpine), **photoperiod** (sunlight: trees grow faster in summer), and **precipitation** (heavy rainfall areas have dense forests).

Types of vegetation

Type	Rainfall	Where	Typical trees and animals
Tropical Evergreen (rain) forests	Over **200 cm**, short dry season	Western Ghats, Lakshadweep, Andaman and Nicobar, upper Assam, Tamil Nadu coast	Trees reach 60 m or more; luxuriant, multi-layered; no definite time to shed leaves, so always green. **Ebony, mahogany, rosewood, rubber, cinchona**. Elephants, monkeys, lemur, one-horned rhinoceros
Tropical Deciduous (monsoon) forests	200–70 cm	Most widespread in India	Trees shed leaves for about **6–8 weeks** in dry summer. **Moist deciduous** (200–100 cm; NE states, foothills, Jharkhand, West Odisha, Chhattisgarh, eastern slopes of Western Ghats): **teak** (most dominant), bamboo, sal, shisham, sandalwood, khair, kusum, arjun, mulberry. **Dry deciduous** (100–70 cm; rainier parts of the plateau and plains of Bihar and UP): teak, sal, peepal, neem. Lion, tiger, pig, deer, elephant
Tropical Thorn forests and scrubs	Less than **70 cm**	North-west India: semi-arid areas of Gujarat, Rajasthan, MP, Chhattisgarh, UP, Haryana	**Acacias, palms, euphorbias, cacti**; roots go deep to find water; stems succulent to conserve water; leaves thick and small to minimise evaporation. Rats, mice, rabbits, fox, wolf, tiger, lion, wild ass, horses, camels
Montane forests	Vary with altitude	Mountain areas	1,000–2,000 m: wet temperate, evergreen broad-leaf **oak and chestnut**; 1,500–3,000 m: temperate coniferous, **pine, deodar, silver fir, spruce, cedar**;

Type	Rainfall	Where	Typical trees and animals
			higher: temperate grasslands; above 3,600 m: **alpine** vegetation (silver fir, junipers, pine, birch), then **tundra** (mosses, lichens). Kashmir stag, spotted deer, wild sheep, jack rabbit, Tibetan antelope, yak, snow leopard, squirrels, shaggy-horn wild ibex, bear, red panda. Nomadic **Gujjars and Bakarwals** use alpine grasslands for grazing
Mangrove forests	Tidal	Coastal deltas of Ganga, Mahanadi, Krishna, Godavari, Kaveri	Roots submerged under water. **Sundari** trees (hard, durable timber) in the Ganga–Brahmaputra delta; palm, coconut, keora, agar. **Royal Bengal tiger**, turtles, crocodiles, gharials, snakes

Wildlife

India has more than **90,000 animal species**, about **2,000 species of birds** (13% of the world total), **2,546 species of fish** (nearly 12% of the world stock), and about 5–8% of the world's amphibians, reptiles and mammals.

- **Elephants**: hot wet forests of Assam, Karnataka and Kerala.
- **One-horned rhinoceroses**: swampy and marshy lands of **Assam and West Bengal**.
- **Wild ass**: arid areas of the **Rann of Kachchh**; **camels** in the Thar Desert.
- **Indian bison, nilgai (blue bull), chousingha (four-horned antelope), gazel** and different species of deer.
- **India is the only country in the world that has both tigers and lions.** The natural habitat of the **Indian lion** is the **Gir forest in Gujarat**. **Tigers** are found in the forests of MP, the **Sundarbans** of West Bengal and the Himalayan region.

- Himalayan animals: yak, shaggy-horned wild ox, Tibetan antelope, bharal (blue sheep), wild sheep, kiang (Tibetan wild ass), ibex, bear, snow leopard, red panda.
- Birds: peacocks, pheasants, ducks, parakeets, cranes and pigeons.

Threats: hunting by greedy hunters for commercial purposes; pollution from chemicals and industrial waste; acid deposits; introduction of alien species; reckless cutting of forests for agriculture and habitation. About **1,300 plant species are endangered** and **20 species are extinct**.

Conservation

Why conserve? Every species has a role in the ecosystem; conservation keeps the ecological balance, which supports life itself.

Steps taken by the government:

- **18 biosphere reserves** set up (for example, **Sunderbans, Nanda Devi, Gulf of Mannar, Nilgiris** and others such as Nokrek, Great Nicobar, Manas, Simlipal, Dihang-Dibang, Dibru Saikhowa, Agasthyamalai, Kanchenjunga, Pachmarhi, Achanakmar-Amarkantak). Several are included in UNESCO's World Network of Biosphere Reserves.
- Financial and technical help to many **botanical gardens** since 1992.
- **Project Tiger**, **Project Rhino**, **Project Great Indian Bustard** and many other eco-developmental projects.
- **National Parks, wildlife sanctuaries and zoological gardens**: there are about 100+ national parks and 500+ wildlife sanctuaries.
- **Wildlife Protection Act (1972)**.

Common people can help by planting trees (**Vanmahotsava**), not buying products of endangered animals and protecting habitats.

EVERGREEN DOES NOT MEAN NO LEAF FALL

Evergreen trees **do shed leaves**, but at different times, so the forest always looks green. Deciduous trees shed leaves **together** in the dry season.

MAP POINTS

Mark vegetation types (Tropical Evergreen, Deciduous, Thorn, Montane, Mangrove) and national parks and sanctuaries such as **Corbett, Kaziranga, Ranthambore, Shivpuri, Kanha, Simlipal, Manas, Gir, Sundarbans, Periyar, Bandipur, Nagarhole**.

Quick recap

- Natural vegetation depends on relief (land and soil) and climate (temperature, photoperiod, precipitation).
- Five types: tropical evergreen, tropical deciduous (most widespread), thorn and scrub, montane, mangrove.
- India is the only country with both tigers and lions (Gir forest for lions).
- Conservation: biosphere reserves (18), Project Tiger, national parks, sanctuaries, Wildlife Protection Act 1972.

Practice questions

1. The most widespread forests of India are

   (a) Tropical Evergreen forests (b) Tropical Deciduous forests (c) Montane forests (d) Mangrove forests

2. Sundari trees are found in

   (a) Thorn forests of Rajasthan (b) Montane forests of Kashmir (c) Mangrove forests of the Ganga–Brahmaputra delta (d) Evergreen forests of the Western Ghats

3. The natural habitat of the Indian lion is

   (a) Sundarbans (b) Kaziranga (c) Corbett (d) Gir forest

4. Rubber, ebony and mahogany are trees of

   (a) Tropical Evergreen forests (b) Tropical Deciduous forests (c) Thorn forests (d) Alpine vegetation

5. Which forests receive less than 70 cm of rain and have trees with long roots and thick, small leaves?

   (a) Mangrove forests (b) Moist deciduous forests (c) Montane forests (d) Tropical thorn forests and scrubs

6. One-horned rhinoceroses live in

   (a) the Rann of Kachchh (b) swampy lands of Assam and West Bengal (c) the Thar Desert (d) the Nilgiri hills

7. Which of these is a biosphere reserve set up to protect flora and fauna?

   (a) Mirzapur (b) Ambala (c) Farakka (d) Gulf of Mannar

8. Distinguish between Tropical Evergreen and Tropical Deciduous forests.

9. What steps have been taken by the government to protect the flora and fauna of India?

Answers: 1. b 2. c 3. d 4. a 5. d 6. b 7. d 8. see notes 9. see notes